

ASSEMBLY LANGUAGE PROGRAM
- Double Precision Addition -

LDA	AL	/ Load A low
ADD	BL	/ Add B low, carry in E
STA	CL	/ Store in C low
CLA		/ Clear AC
CIL		/ Circulate to bring carry into AC(16)
ADD	AH	/ Add A high and carry
ADD	BH	/ Add B high
STA	CH	/ Store in C high
HLT		

AL,	...	/Location of operands
AH,	...	
BL,	...	
BH,	...	
CL,	...	
CH,	...	



ASSEMBLY LANGUAGE PROGRAM

- Logic and Shift Operations -

- Logic operations

- BC instructions : AND, CMA, CLA
- Program for OR operation

LDA	A	/ Load 1st operand
CMA		/ Complement to get A'
STA	TMP	/ Store in a temporary location
LDA	B	/ Load 2nd operand B
CMA		/ Complement to get B'
AND	TMP	/ AND with A' to get A' AND B'
CMA		/ Complement again to get A OR B

- Shift operations - BC has *Circular Shift* only

- Logical shift-right operation - Logical shift-left operation

CLE
CIR

CLE
CIL

- Arithmetic right-shift operation

CLE	/ Clear E to 0
SPA	/ Skip if AC is positive
CME	/ AC is negative
CIR	/ Circulate E and AC



SUBROUTINES

Subroutine

- A set of common instructions that can be used in a program many times.
- Subroutine linkage : a procedure for branching to a subroutine and returning to the main program

Example

Loc.			
		ORG 100	/ Main program
100		LDA X	/ Load X
101		BSA SH4	/ Branch to subroutine
102		STA X	/ Store shifted number
103		LDA Y	/ Load Y
104		BSA SH4	/ Branch to subroutine again
105		STA Y	/ Store shifted number
106		HLT	
107	X,	HEX 1234	
108	Y,	HEX 4321	
			/ Subroutine to shift left 4 times
109	SH4,	HEX 0	/ Store return address here
10A		CIL	/ Circulate left once
10B		CIL	
10C		CIL	
10D		CIL	/ Circulate left fourth time
10E		AND MSK	/ Set AC(13-16) to zero
10F		BUN SH4 I	/ Return to main program
110	MSK,	HEX FFF0	/ Mask operand
		END	

